# ERFAN HOSSEINI SERESHGI

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## PROFESSIONAL EXPERIENCE

**Postdoctoral Fellow, The Murphy Institute, Tulane University** — **Aug 2025 – present**
*New Orleans, LA*

**Graduate Research Assistant, Tulane University** — **Sep 2021 – May 2025**
*New Orleans, LA*

**Data Scientist, Tulane University (LA-CEAL NIH Project)** — **Jan 2021 – Nov 2021**
*New Orleans, LA*

**Research Software Engineer, Tulane School of Public Health** — **Jan 2021 – Nov 2021**
*New Orleans, LA*

**IT Specialist, Tulane Pre-college Programs** — **Summer 2020**
*New Orleans, LA*

**Jr. Product Manager, AIESEC** — **Aug 2017 – Feb 2018**
*Tehran, IR*

**Front-end Developer Intern, Moduland.ir** — **Summer 2017**
*Tehran, IR*

## EDUCATION

**Tulane University – Ph.D. in Computer Science** — **2018 – 2025**
*New Orleans, LA*

**Amirkabir University of Technology – B.Sc. in Computer Science** — **2014 – 2018**
*Tehran, IR*

## RESEARCH AND PUBLICATIONS

**Semantic Anomaly Detection in Human Trajectories: Preserving Behavioral Patterns Through Calendar Representations** — **2025**
*(E. Hosseini Sereshgi, M. Uppalapati, Y. Liu, L. Kennedy, A. Züfle, C. Wenk),*
The 2nd ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection, 10 pages. (just accepted)

**Drawing Reeb Graphs** — **2025**
*(E. Chambers, B.T. Fasy, E. Hosseini Sereshgi, M. Löffler),*
*The* 36th International Workshop on Combinatorial Algorithms (IWOCA 2025)

**MAGIC: Map And Geographic Information Construction, Comparison and Conflation** — **2025**
*(E. Hosseini Sereshgi),*
Ph.D. Dissertation, Tulane University

**ROADSTER: Improved Algorithms for Subtrajectory Clustering and Map Construction** **2025**

*(K. Buchin, M. Buchin, J. Gudmundsson, J. Hendriks, E. Hosseini Sereshgi, R. I. Silveira, J. Sleijster*, F. *Staals, C. Wenk),*
Computers and Geosciences Journal, Vol. 196, 49 pages.

**Length-preserving Matching for Closed Curves** **2024**

*(E. Hosseini Sereshgi, M. Löffler*, F. *Staals, C. Wenk),*
*The* 31st Fall Workshop on Computational Geometry, 5 pages.

**Map Stitcher: Graph Sampling-based Map Conflation** **2024**

*(E. Hosseini Sereshgi, and C. Wenk),*
*The* 3rd ACM SIGSPATIAL International Workshop on Spatial Big Data and AI for Industrial Applications (GeoIndustry), 11 pages.

**Graph Sampling for Map Comparison** **2024**

*(J. Aguilar, K. Buchin, M. Buchin, E. Hosseini Sereshgi, R.I. Silveira, C. Wenk),*
ACM Transactions on Spatial Algorithms and Systems 10(3), 24 pages.

**Drawing Reeb Graphs** **2023**

*(E. Chambers, B.T. Fasy, E. Hosseini Sereshgi, M. Löffler, S. Percival),*
*The* 31st International Symposium on Graph Drawing and Network Visualization, Poster paper.

**On Length-sensitive Fréchet Similarity** **2023**

*(K. Buchin, B.T. Fasy, E. Hosseini Sereshgi, C. Wenk),*
The 18th Algorithms and Data Structures Symposium (WADS): 208-231.

**Merging Roadmaps using Graph Distance Measures** **2022**

*(E. Hosseini Sereshgi and C.Wenk),*
The 30th Fall Workshop on Computational Geometry, 5 pages.

**Graph Sampling for Map Comparison** (received best paper award) **2021**

*(J. Aguilar, K. Buchin, M. Buchin, E. Hosseini Sereshgi, R.I. Silveira, C. Wenk),*
*The* 3rd ACM SIGSPATIAL International Workshop on Spatial Gems, 8 pages.

**Measuring Length-Preserving Fréchet Correspondence for Graphs in $R^2$** **2021**

*(K. Buchin, B. T. Fasy, E. Hosseini Sereshgi and C. Wenk),*

*The* 29th Fall Workshop on Computational Geometry, 4 pages.

**Improved Map Construction using Subtrajectory Clustering** **2020**

*(K. Buchin, M. Buchin, J. Gudmundsson, J. Hendriks, E. Hosseini Sereshgi, V. Sacristán, R. Silveira, J. Sleijster, F. Staals, C. Wenk),*
*The* 4th ACM SIGSPATIAL Workshop on Location-Based Recommendations, Geosocial Networks, and Geoadvertising: article 5, 4 pages.

**Computing Relevant Subtrajectory Bundles Faster** **2020**

*(E. Hosseini Sereshgi and C. Wenk),*
SoCG, Computational Geometry: Young Researchers Forum, 2 pages.

**Clustering Gene Expression with Polygonal Chain Alignment** **2018**

*(H. Banaee, E. Hosseini Sereshgi, A. Mohades, F. Zare)*
*B.Sc. capstone project, Amirkabir University of Technology*

## PRESENTATIONS

**Length-preserving Matching for Closed Curves** **2024**
31st Fall Workshop on Computational Geometry (FWCG)

**Map Stitcher: Graph Sampling-based Map Conflation** **2024**
3rd ACM SIGSPATIAL International Workshop on Spatial Big Data and AI for Industrial Applications (GeoIndustry)

**On Length-sensitive Fréchet Similarity** **2023**
18th *Algorithms and Data Structures Symposium (WADS)*

**Merging Roadmaps using Graph Distance Measures** **2022**
30th *Fall Workshop on Computational Geometry (FWCG)*

**Graph Sampling for Map Comparison** **2021**
3rd *ACM SIGSPATIAL International Workshop on Spatial Gems*

**Measuring Length-Preserving Fréchet Correspondence for Graphs in $\mathbb{R}^2$** **2021**
*29th Fall Workshop on Computational Geometry (FWCG)*

**The Study of Gentrification on Social Urban Simulation - How Income and Interest Can Shape Neighborhoods** **2020**
*Tulane University*

**Improved Map Construction using Subtrajectory Clustering** **2020**
*4th ACM SIGSPATIAL, Location-based Recommendations, Geosocial Networks and Geoadvertising*

**Computing Relevant Subtrajectory Bundles Faster** **2020**
*36th International Symposium on Computational Geometry (SoCG), Young Researchers Forum*

**Clustering Gene Expression with Polygonal Chain Alignment** **2018**
*Amirkabir University of Technology*

**A brief Intro to Computational Geometry** **2017**
*Amirkabir University of Technology, Graduate studies seminar*

## TEACHING EXPERIENCE

**Algorithms (Graduate level)** **Fall 2024**
*Teaching assistant*

**Computational Geometry (Graduate level)** **Fall 2023**
*Teaching assistant*

**Arduino course at Tulane Pre-college Program** **Summer 2022**
*Instructor*

**Electrical Engineering course at Tulane Pre-college Program** **Summer 2022**
*Instructor*

**Introduction to Discrete Math** **Fall 2020**
*Teaching assistant*

**Introduction to Algorithms** **Fall 2019**
*Teaching assistant*

**Python game design at Tulane Pre-college Program** **Summer 2019 and 2022**
*Instructor*

**Robotics at Tulane Pre-college Program** **Summer 2019 and 2022**
*Instructor*

**Intro to Computer Science 1 (Python)** **Fall 2018, Spring 2019, Spring 2020**
*Teaching assistant*

**Operating systems Lab/Workshop** **Spring 2017**
*Teaching assistant*

**C++ Programming teacher at Helli 4 high school** **2014 – 2015**
*Instructor*

## LEADERSHIP & VOLUNTEER EXPERIENCE

- Tulane Computer Science Graduate Student Council (CSGSC) founding member
- Tulane Computer Science Graduate Student Council department representative (Spring 2023)
- Tulane Graduate Studies Student Association Travel fund committee member (2023)
- Senator at Tulane Graduate and Professional Student Association (2020-2022)
- Tulane Graduate Studies Student Association representative (2020-2021)
- IT team leader at AIESEC in University of Tehran (2017-2018)
- Marketing designer at AIESEC in Amirkabir University of Technology (Spring 2017)
- AIESEC global volunteer for raising public awareness about clean energy and recycling in Guangzhou, China. (Summer 2016)
- Member of scientific association of math and computer science at Amirkabir University of Technology (2015-2016)

## CERTIFICATES

- Fundamentals of Deep Learning from Nvidia
- Java programming from Amirkabir University of Technology
- Android development from Amirkabir University of Technology
- Web development and web design from Amirkabir University of Technology
- CITI Group1. Biomedical Researchers
- CITI Group4. IRB, Biomedical Research

## SKILLS & ABILITIES

- Skilled in Python, C++ and Java
- Familiar with HTML, CSS, Javascript and SQL
- Have worked with R and R studio
- Familiar with Git and Visual Paradigm
- Have some intermediate knowledge about Android Studio

- Have experience working with Adobe Photoshop and illustrator
- Familiar with QGIS, ArcGIS and GDAL libraries
- Familiar with Pytorch, Keras and Scikit-Learn

## LANGUAGES

- English (fluent)
- Persian (native)
- Arabic (intermediate)

## HONORS & AWARDS

- Tulane Connolly Alexander Institute for Data Science Summer Graduate Award (2023)
- Best paper award at ACM SIGSPATIAL: Spatial Gems (2021)
- Ranked 6$^{th}$ in the Iranian national CS graduate school entrance exam (2018)
- Ranked among top 5 computer science students at Amirkabir University of Technology (class of 2018)
- Semi-finalist in 2014 BAYAN coding contest in Iran
- Ranked among 3% in the Iranian national university/college entrance exam (2014) (More than 60,000 students)
- Was selected by and studied at the national organization for development of exceptional talents (NODET) in Iran

## ACADEMIC SERVICES

- Program committee at the 2nd ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection
- Reviewer at the 33rd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems
- Reviewer for the International Journal of Geographic Information Systems
- Reviewer for the International Colloquium on Automata, Languages, and Programming
- Reviewer at the 32nd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems
- Reviewer at the 29th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems
- Reviewer at the 16th International Symposium on Spatial and Temporal Databases
- Reviewer at the 28th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems
- Reviewer at the 17th Algorithms and Data Structures Symposium (WADS21)